Qns	Answer	Marks
4 (a)(i)	The Principle of Conservation of Linear Momentum states that the **total linear momentum** of an **isolated system of interacting bodies** before and after collision remains constant if **no net external force acts on the system**. OR The Principle of Conservation of Linear Momentum states that <u>the total linear momentum of a system remains constant provided that no resultant external force acts on the system</u>.	**B1**
(a)(ii)	The **resultant force acting on a body** is **proportional to the rate of change of momentum** of that body.	**B1**
(b)(i)	After collision, the bodies will move along the same line that joins the 2 bodies' centre of mass rightwards.	**B1**
(b)(ii)	By principle of conservation of momentum: $mu + 14m(0) = 15mV$ $V = \frac{1}{15}u$ Ratio of kinetic energy $= \frac{KE_{nitrogen}}{KE_{neutron}} = \frac{\frac{1}{2}MV^2}{\frac{1}{2}mv^2} = \frac{15\cancel{m}\left(\frac{\cancel{v}}{15}\right)^2}{\cancel{m}\,\cancel{v^2}} = \frac{1}{15} = 0.067$	**C1** **M1** **A1**
(b)(iii)	From (bii), since initial total KE before collision not equals to final total KE after collision, the collision is not elastic (inelastic). OR Since there is no relative separation between the neutron and nitrogen atom after collision, the relative speed of approach is not equals to relative speed of separation, the collision is not elastic (inelastic).	**M1** **A1**

Qns	Answer	Marks
5 (a)	When S is opened, current in coil Y stops flowing and the magnetic flux	B1